

Air Visio: Air Quality Monitoring and Analysis Based Predictive System

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Abstract— Sri Lanka is facing a serious air pollution problem that severely impacts the daily life of every Sri Lankan. The main source of ambient air pollution in Sri Lanka is vehicular emissions. A methodology to monitor the air quality in real-time with an overall coverage of Sri Lanka, and automatically process these huge data to identify air quality levels in a specific area is now becoming a timely research topic. An air quality monitoring and analysis based predictive system is proposed to monitor the ambient air quality, provides the best route with minimum polluted air, maps the heatmaps to identify the current air quality of an area easily and predict the future air quality of each area. The prototype was implemented by hierarchically deploying two different gas sensors, an Arduino Uno board and a wifi module, to implement in open spaces between smart buildings, and transfers the sensor data back to the information processing center by using IoT technology for real-time display. The information processing center stores real-time information which is collected from the sensors to the database. By reading sensor data stored in the database, the front-end system draws real-time, accurate air quality levels included maps and predicts the less polluted routes and the air quality level over an area. Further, an energy harvesting system is also presented for the power consumption of the device. A route is suggested in an accuracy of 70% from this system. The final product provides a low cost, highly portable and easily maintainable system for the users.

Keywords— Air quality monitoring, Internet of Things, heat map generating, route suggestion, air pollution prediction

I. INTRODUCTION

Both indoor and outdoor air pollution has grown into an alarming state all over the world. Although air pollutants are many, the most important are Particulate Matter (PM), Carbon Dioxide (CO₂), Carbon Monoxide (CO), Ground-level Ozone (O₃), Nitrogen Oxides (NO_x), Sulfur Oxides (SO_x) and Lead (Pb) which is found in the ambient air. Now it has begun a major environmental health problem that affects people in both developed and developing countries.

When considering outdoor air pollution, emissions caused by the combustion process from motor vehicles, solid fuel burning and industry are the common sources of outdoor air pollution. In 2012 alone, approximately 24,000 people died of air pollution in Sri Lanka [1]. Respiratory health symptoms which cause due to outdoor air pollution

are, upper and lower respiratory tract infections, pneumonia, bronchitis, asthma, poor lung functions, reduce lung growth and lung cancer [2]. Ambient (outdoor) air pollution in both cities and rural areas was estimated to cause 4.2 million premature deaths worldwide per year in 2016 [3]. The main source of ambient air pollution in Sri Lanka is vehicular emissions, which contribute to over 60% of the total emission in Colombo [4]. According to the population in Sri Lanka, school children are the most effective group which exposed to high levels of pollutants exaggerated by overcrowding of major schools in cities, especially in Colombo and Kandy. Air pollutant levels were significantly higher in the premises of the urban school as compared to the rural school [5].

To address this problem, this research proposes a prototype which uses an air quality monitoring device, energy harvesting mechanism and a mobile application to visualize air quality information in terms of route suggestions, heat map, and air quality predictions. People's awareness is the main objective of this research and it helps people who are suffering from respiratory diseases, kids and older persons to avoid the most polluted areas by identifying highly polluted areas when traveling.

II. LITREATURE REVIEW

There are a number of research projects that have been conducted to measure air quality based on IoT or wireless sensor networks. National Institute of Engineering in Mysuru, India developed an IoT system that collects real-time air pollution data in specific locations[6]. Mapua University in Philippine created a prototype that detects air pollution and particulate matter. The designed hardware sends an email notification to the registered email addresses whenever the sensor detects high reading on either the particulate matter or Carbon monoxide [7]. Devi Ahilya University in India developed an air quality monitoring system and it uses low-cost air quality monitoring nodes comprises of semiconductor sensors to measure CO, CO₂, SO₂ and NO₂ with the Wi-Fi module [8]. The National Policy on Urban Air Quality Management was adopted in 2000 in order to reduce outdoor air pollution due to vehicular emissions in Sri Lanka. Banning the importation of two-stroke three-wheelers in 2008, initiation of vehicular emission testing (VET) program in 2008 the introduction of low sulfur

diesel in 2003 are some key steps that have been taken to reduce ambient air pollution in Sri Lanka. With the publication of WHO air quality guidelines in 2005 [9], air quality standards for Sri Lanka (PM₁₀ and PM_{2.5}) were amended and gazette on August 2008 [10].

Many organizations in Sri Lanka, such as Central Environmental Authority (CEA), National Building Research Organization (NBRO), Department of Motor Traffic (DMT), etc. monitor the ambient air quality in Sri Lanka. The Central Environmental Authority has deployed two fixed automated stations deployed in Battaramulla and Kandy, to monitor the ambient air quality continuously. The mobile station is currently in progress state. Apart from this method, the most used monitoring method is the passive monitoring network which was developed by the National Building Research Organization and also they are currently working on a sensor deployment to monitor ambient air pollution. Therefore, the current existing air quality monitoring system in Sri Lanka is still in a research state and need an improvement to increase usability.

III. SYSTEM IMPLEMENTATION

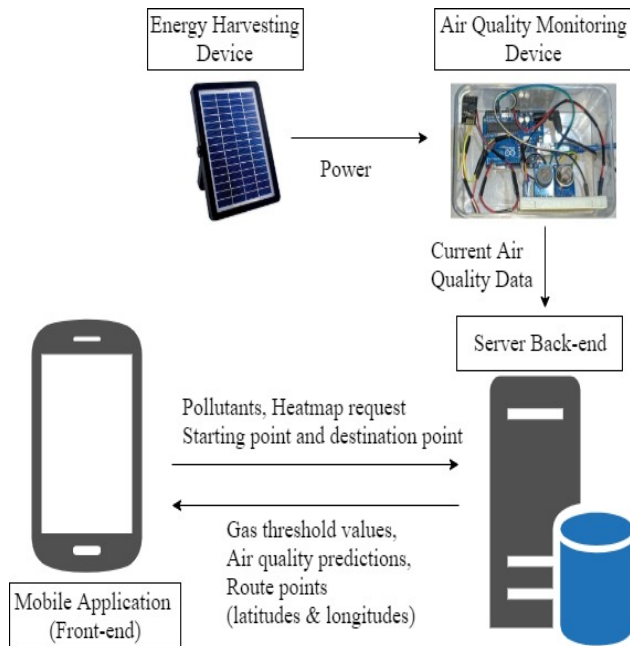


Fig. 1. System hierarchy

As Fig.1 shows, the whole system consists of three main parts. The first part is the IoT based air quality monitoring hardware system, the second part is the server backend and the third part is the front-end mobile application. This IoT based air quality monitoring hardware system is fixed at suitable urban, open areas to monitor the ambient air quality of these regions. This hardware system uses an Arduino Uno board and esp8266 Wi-Fi module to transmit the information packet to the MongoDB database. This hardware system also consists of an energy harvesting device to obtain the required power to operate. The mobile application gets real-time air quality information by accessing the database and displays the best route with minimum polluted air, maps the heat maps to identify the

current air quality of an area easily and predicts the future air quality of each area.

A. Air Quality Monitoring Device

In this component, MQ4, and MQ7 sensors were used to measure Methane (CH₄) and Carbon Monoxide (CO) gases in the air respectively. The conductivity increases with the increment of gas concentration in the polluted air, when polluted air goes through these sensors. The difference of conductivity can be converted into gas concentration values, by implementing an Arduino code using the Arduino Uno board. The output analog value (0 - 4095) of each sensor should be converted into sensor voltage value using the (1).

$$\text{Sensor Voltage} = (\text{Analog Reading} * 3.3\text{V}) / 4095 \quad (1)$$

After sensor voltage calculation, we can convert that value into a Parts per Million (PPM) concentration value using the sensitivity calibration curve of each gas sensor given in their datasheets separately, which were drawn under the standards between the two axis; the gas concentration (PPM) and sensor voltage (V). With the aid of those sensitivity curves of each gas sensor, built two relationships as (2) and (3) for the sensors separately, to convert the sensor voltages into concentration value. Using the graphical analysis software, Engauge Digitizer; points of these graphs were selected and drew an exponential to that graph using the selected points. With the aid of that drawn exponential, declared equations to each graph for building the relationship between sensor voltage and gas concentration in ppm. The equations built for the relationship of gas concentration (ppm) and sensor voltage (V_{RL}) of two sensors are as follows.

$$\text{CH}_4 \text{ concentration} = 10.938 * e^{(1.7742 * V_{RL})} \quad (2)$$

$$\text{CO concentration} = 3.027 * e^{(1.0698 * V_{RL})} \quad (3)$$

The ESP8266 Wi-Fi module is used to transmit the sensor data to the database in every one minute of time interval to make the device more real in time. The location of the device is identified using a predefined terminal number to the places where the devices implemented. Instead of using a GPS system, this predefined terminal number system was used for reducing the size of the data packet, as this system is a more real-time system and it uploads the data in every 1 minute of the time interval. This allocated terminal number is also uploaded with the sensor data for the ease of identification of device location.

The data packet is sent by using a Hypertext Transfer Protocol (HTTP) Post request. The packet uses encoded format when sending to the server, to increase the security of the data packet. A Transmission Control Protocol (TCP) connection is established with the server by the Wi-Fi module. An Arduino Uno board is used to implement the data processing code. This Arduino code handles the sensors to get the sensor inputs, process them and handles the Wi-Fi module to send these processed outputs to the database.

B. Energy Harvesting Device

Energy harvesting is the process of collecting, wasted environmental energy lost as heat, light, sound, vibration or movement and then converting that energy into small units of useful electric power. Among the various energy conversion methods most widely used method is solar energy harvesting (SEH). It has been widely shown as the greatest potential to achieve better performance in higher voltage and power density levels. The prototype designed in this paper is based on solar electric method. Fig. 2 gives a high level overview of the SEH system. The system mainly consists of 2 parts: Harvesting Circuit and Timer Circuit.

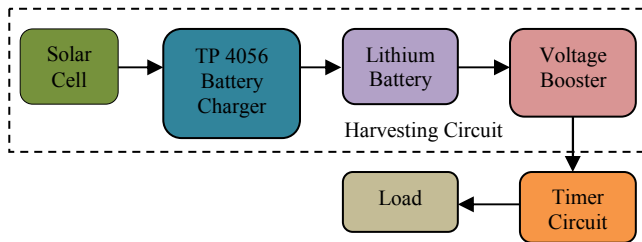


Fig. 2. A SEH system

Solar Energy Harvesting (SEH) circuit was implemented to self-charge (user free energy from the solar cell) the batteries, even put Arduino to sleep to save power on Arduino boards. This circuit can solve the issue in, even when the Arduino is sleeping, and the sensor might still be active, draining batteries. This uses some piezo elements to charge the batteries, and boost the voltage from it to the 5v used by the Arduino Uno. Harvesting circuit consists of solar cells, TP4056 battery charger, and voltage booster. Solar power converts energy from sunlight into electricity. Electric power generated by solar cells, input to the lithium battery charger (TP4056). The output of the battery charger is stored in the lithium battery. This module contains a separate protection IC and TP4056 Li-Ion charge controller IC. The main benefits of the TP4056 battery charger is it can control the trickle charge, charging at a constant voltage of a connected lithium battery, short-circuit protection and over current, over-discharge protection and overcharge protection. TP4056 module Includes two indicator LEDs (Red and Blue) to depict charging and whether it is charged completely. A voltage step-up booster, which is connected to the battery, is used to convert 3.7V DC to 5V DC. Timer circuit which was shown in Fig. 3 was added to the piezo charged battery which powers Arduino only for a few seconds and then turns it off again to save power. It fits applications where the microcontroller is only used to read sensors, transmit or save the data, and go back to sleep for a few minutes.

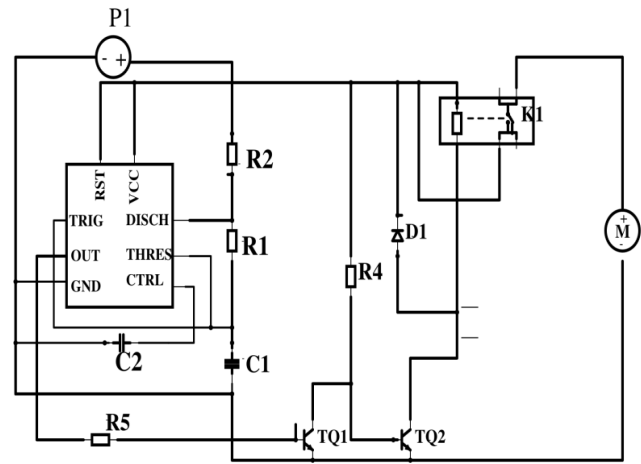


Fig. 3. Timer circuit

C. Server Back-End

The server back-end is the most important part of the research, which is responsible for data storing and processing. MongoDB is the database of the Air Visio system, it is a cross-platform, open source and document-oriented database. The sensor data received from the air quality monitoring device are processed and stored in the database but that sensor data cannot be stored in the database directly. So the server back-end must be built using NodeJs language and then the sensor data passes through the server back-end to the database. The NodeJs is primarily used for non-blocking event-driven servers, due to its single threaded nature. As well as the server backend with the database hosted in a server to receive the data packet from air quality monitoring device and the data packet travel through the internet to the server back-end.

This data packet contains three values; methane (CH_4) and carbon monoxide (CO) gases and terminal number. The terminal number is like a two-digit number. It is used to identify the location of air quality monitoring device.

D. Heat Map Generation

A heat map is a graphical representation of data that uses a system of color-coding to represent different values. It used in various forms of analytics but is most commonly used to show user behavior on specific situation. There are two heat maps were generated for methane and carbon monoxide separately. It is generated using accurate sensor data stored in the database, JavaScript and HTML language. Latitude and longitude are related to the terminal number, which is used to identify the location. Then using color schemes, the sensor values related to the location are displayed separately in two maps. Moreover, it displays the real time level of air pollution in an area. Many different color schemes can be used to illustrate the heat maps.

Methane gas concentration threshold ranges can be divided into four levels and it is given in TABLE I.

TABLE I. NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH) STANDARD FOR METHANE GAS [11]

Gas concen tration (PPM)	Description	Interpret ation	Color Code
0-50	Air pollution is little or no risk.	Good	Green
51-70	Air quality is acceptable, but some people who are very sensitive to air pollution can be affected.	Moderate	Yellow
71-150	Everyone may begin to affect health effects.	Unhealthy	Red
151-500	Everyone may be affected by serious health effects.	Hazardous	Maroon

Carbon Monoxide gas concentration threshold ranges can be divided into four levels and it is given in TABLE II.

TABLE II. OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA) STANDARD FOR CARBON MONOXIDE GAS [12]

Gas concen tration (PPM)	Description	Interpretat ion	Color Code
0-1000	Air pollution is little or no risk.	Good	Green
1001-49999	Everyone may begin to affect health effects.	Unhealthy	Red
50000-150000	Methane would explode and cause deaths to everyone around the area.	Potentially explosive	Dark red
Over 500000	Methane would displace oxygen in confined areas, resulting in an oxygen-deficient atmosphere and displace of oxygen in the blood.	Hazardous	Maroon

In the mobile application, users can use the heat maps to identify the air polluted in an area, due to high amounts of various gases in that area. The processed heat map is shown in Fig. 4.

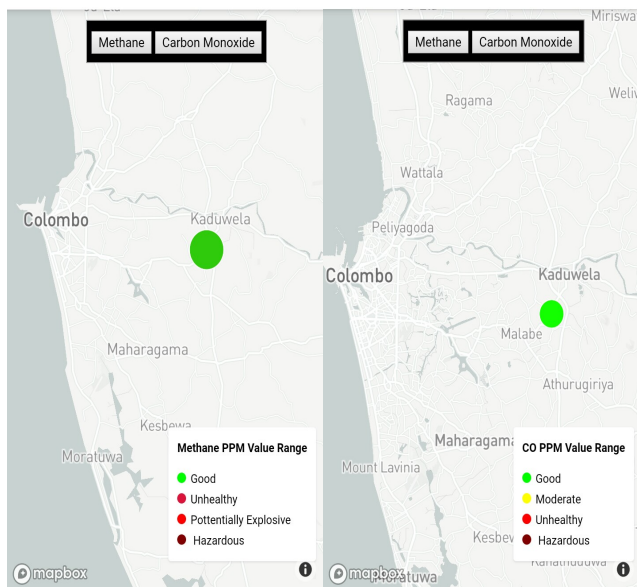


Fig. 4. Heat Map visualization in mobile application

E. Mobile App

‘AirVisio’ is an android based mobile application that will be used to find the best route from one place to another place, with a minimum amount of polluted air, identify current air quality level at any location via a heatmap and predict future air quality levels. The Mobile app includes a Splash screen and an onboard screen to make it more attractive for the users. It uses a Map SDK to load the Google map, a place API to provide autocomplete place suggestions and a direction API to draw the route between two points. This application connects with a back-end server to get air pollution details to process heatmap, air quality predictions and route suggestions.

F. Route Suggestion Process

The system suggests the best route for the client based on the air quality and the distance for a given destination. A sample sketch that used to suggest routes is shown in Fig. 5.

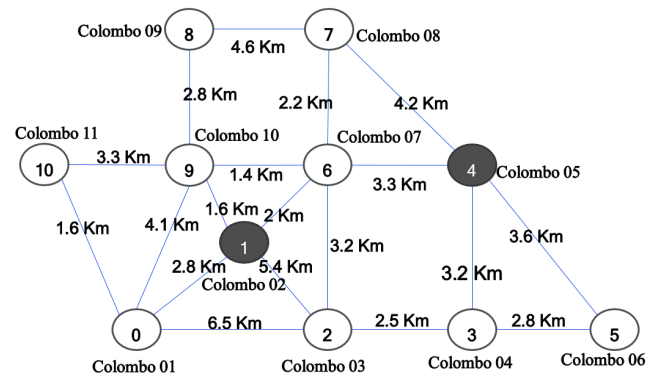


Fig. 5. Sample sketch of all sensor nodes

These nodes represent 11 cities within the Colombo district, Sri Lanka. Before suggesting routes, the system filters the minimum polluted cities by looking at the active air quality records for the above-given cities. If Node1, Node3, and Node10 are polluted, the route suggestion process will dynamically create a map by removing all polluted points that filtered as shown in Fig. 6.

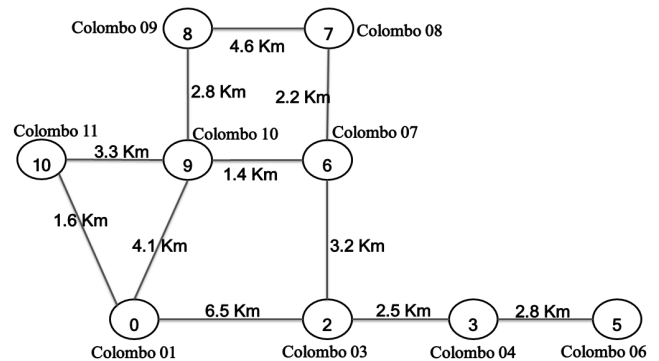


Fig. 6. Sample sketch of dynamic map

A new algorithm has been developed by combining Dijkstra’s algorithm and the dynamic mapping process to suggest the best way. The algorithm takes the nearest non-polluted point for the given start as the starting point and

the nearest non-polluted point for the given destination as the endpoint for real-life applications. Finally, the suggested points for the route are sent to the Mobile application to display it on the map. When displaying the best route on the map, the red marker indicates the starting point and the blue marker indicates the endpoint of a route. Furthermore, the green marker displays the non-polluted points within the route. The comparison with default Google map is shown in Fig. 7 and Fig. 8.

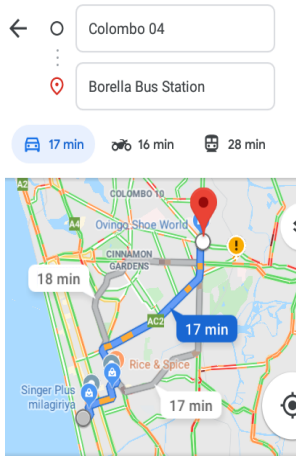


Fig. 7. Route visualization without considering air pollution

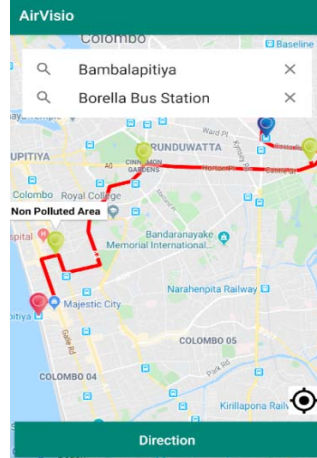


Fig. 8. Minimum polluted and accurate route

G. Air Quality Index Prediction

The original air pollution data (NO_2 , SO_2) are from the passive sampling stations of the National Building Research Organization and Central Environmental Authority, collected on an Hourly basis over a period of 20 years. There are some missing data in the original data set, because of the power failures, A/C failures, system failures, and calibration failures. This type of missing data called Missing Completely At Random (MCAR). It occurs when the probability of missing value depends neither on the variable itself nor of another variable of the database; in mathematical terms, it is written:

$$P(r | X_{\text{obs}}, X_{\text{mis}}) = p(r) \quad (4)$$

where X_{obs} is the observed data, X_{mis} is the MD and (r) is the distribution condition of MD[13]. There were 12.5% of missing data and the summary is given with the TABLE III.

TABLE III. SUMMARY OF MISSING DATA

	Missing		Valid N	Mean	Std. Deviation
	N	Percent			
SO ₂	30	12.5%	210	11.89954	30.261309
NO ₂	30	12.5%	210	9.93800	15.581705

In this paper, multiple imputation method is used to deal with missing data. Multiple imputation method is instead of substituting a single value for each missing data, the missing values are replaced with a set of colorable values predicted with the natural uncertainty of the right values and variability.

This paper presents with seven models, SARIMA model, Naive, Simple average, moving average, simple exponential smoothing, holt's linear trend method, holt

winters method and LSTM (Long-Short-Term-Memory) using the location of 6.901009(North) and 79.92682(East), respectively.

TABLE IV. RMSE VALUES

Model	RMSE value
SARIMA	6.03
LSTM	15.94
Naive	20.10
Sample Average	13.15
Moving Average	8.92
Simple exponential smoothing	8.99
Holt's linear trend method	14.73
Holt winters method	9.56

According to the RMSE values of each model, Seasonal ARIMA (SARIMA) are selected for model development. This prediction uses the monthly means of the data, for the air quality index prediction. Data from 2012 to 2018 are used for training, while data in 2018 is used for testing.

In model development, the first step is time series identification by plotting sample autocorrelations (SAC) and sample partial autocorrelations (SPAC) based on the original data. Then the stationarity of the data should be identified and outliers should be removed. The observations of correlograms indicate that more spikes are found in the regular differencing and seasonal differencing. Then, the regular and seasonal differences are chosen for the model development. It indicates that the selected model was fit. Before starting the model development, the AQI value of each air pollutant was calculated. The final SARIMA model (3,1,1) (3,1,1,12) was used to forecast the values of the 12 months-ahead.

IV. TEST RESULTS

Powered on the Air Visio device using the energy harvesting device and the data collecting, processing and sending steps were gradually took place. For the database management, we used mongo dB database. This will handle all the data transactions in the system. The device with the Carbon monoxide gas sensor and the methane gas sensor collects air quality data and send it to the database once per every one minute as a data packet. Then the data packets received to the database were visualized in a heat map as above. The two measured gas concentrations were displayed in the given specific colors accordingly the range of each gas's concentration level as shown in Fig. 9.

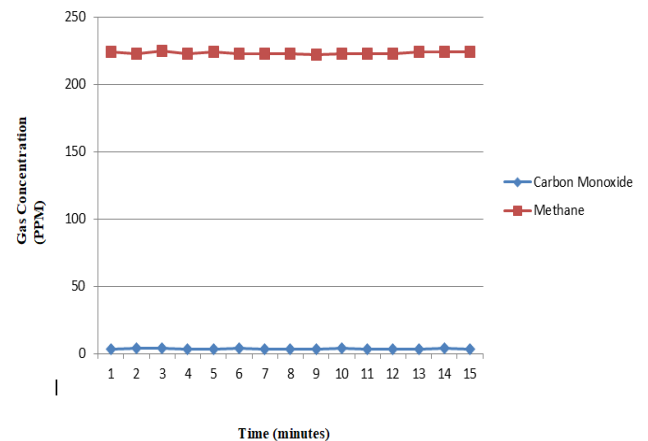


Fig. 9. Behavior of the CO and Methane gases within 15 minutes

Error measuring metrics represent the accuracy of the forecasting. It is obtained by comparing the forecast values and the original data. In this paper, Mean Absolute Error was used as the error measuring metric. The result showed MAE value for the selected model was 6.03. Thus, the empirical results indicated that the model was able to accurately represent the Air pollution historical data set. Prediction was tested for four locations.

TABLE IV. SAMPLE OF FORECAST DATA (AQI FOR NO₂-NI)

Location-CEA			
Month	Predicted	Actual	Error
2018-01-31	49.641655	50.00	0.358345
2018-02-28	36.658035	32.50	4.158035
2018-03-31	32.891162	37.50	4.608838
2018-04-30	18.254488	20.00	1.745512
2018-05-31	18.763705	22.50	3.736295
2018-06-30	16.709432	20.75	4.040568
2018-07-31	25.883197	21.25	4.633197
2018-08-31	22.814599	25.00	2.185401
2018-09-30	28.886280	28.75	0.13628
2018-10-31	39.790173	45.00	5.209827
2018-11-30	39.185702	43.00	3.814298

The Route suggestion process has been tested for nodes 0 to 2 and the results as shown in Fig. 10.

Dynamic map points "response": [	Latitude and longitude values for best route based on air pollution,
{ "start": 0,	"location_path": [
"end": 4,	{ "lat": 6.9361,
"distance": 1 },	"lon": 79.845 },
{ "start": 4,	{ "lat": 6.8741,
"end": 3,	"lon": 79.8605 },
"distance": 2 },	{ "lat": 6.901,
{ "start": 3,	"lon": 79.8552 },
"end": 2,	{ "lat": 6.9128,
"distance": 1 }],	"lon": 79.8507 }]

Fig. 10. Test results from route suggestion algorithm

V. CONCLUSION

This paper has presented a comprehensive approach to detect the ambient air quality level and predict the pollution levels of each area and suggest the best route with minimum polluted air to travel.

Air quality monitoring device based on IoT technology is used to detect Carbon monoxide and methane gas concentration values in each area and send these real-time sensor data to the database in a data packet including two concentration values and the location terminal number of the device in every one minute of the time interval. It is smaller in size, low in cost, easily maintainable, portable and can be fixed at any location according to the requirements. The power required to operate the device will

be given using the energy harvesting device created by piezoelectric sensors. Mobile application users are able to use this Air Visio mobile application to find the best route with minimum polluted air and check the current air quality level at any location via a heat map. Also, users can check the future air quality level of any selected location, using the air quality prediction service in this application.

Air Visio device can be improved using more powerful sensors to measure other gases included in ambient air, instead of MQ4 and MQ7 sensors. Then the heatmap generation and future air quality prediction processes can also be improved to a further extent.

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